PRINCIPLES OF FISH BIOLOGY (ZOO603T)

CURRENT FINALTERM PAPER SPRING 2024(27JUNE)

SHORT QUESTIONS(3)

1. Edible fishes name

- Salmon
- Crab
- Lobster
- Trout
- Rohu
- Herring
- Shrimp

2. Three parts of fish brain

Fish brains are generally divided into general regions. •

1) Prosencephalon,
2) Mesencephalon and
3) rhombencephalon are the regions of brain.

3. Types Of Scales

There are four main kinds of scales and numerous variations of each kind.

(i) Placoid (sharks and rays)
(ii) Cosmoid (lungfishes and some fossil fishes)
(iii) Ganoid (bichirs , Bowfin, paddlefishes, gars, sturgeons)
(iv) Cycloid and Ctenoid (most bony fishes)

4. Which character include in fish morphology

Specific characters help identify and classify fish species, and understand their evolution, adaptation, and diversity. These are;

- Body shape
- Mouth type
- Mouth location
- Scales
- Gills
- Fin type
- Barbels
- Lateral line system

5. Food web:

A network of food chains showing feeding relationships between species in an ecosystem, illustrating energy flow from producers to consumers and decomposers.

6. Jugular Fin

When pelvic fins are below the pectoral fins, they termed thoracic. When a thoracic pelvic fin is attached under gills, it may called jugular.

7. Name of causative agent of furunculoris and abdominal

Abdominal dropsy of Carps is caused by **Aeromonas punctuata.**

Furunculoris of Salmons and trout's is caused by **Aeromonas salminicida.**

8. What is Pisciculture?

In pisciculture (culture fisheries), which generally pertains to small water bodies, the fish seed has to be sown, tended, nursed, reared and finally harvested when grown to table size.

9. Green revolution& Blue revolution

"BLUE REVOLUTION" (fish production) on the same lines

A period of rapid growth in aquaculture, achieved through the development of new technologies, feeds, and management practices, aimed at increasing seafood production.

'GREEN REVOLUTION' (for producing enough food for all).

A period of significant increase in agricultural productivity, achieved through the use of high-yielding crop varieties, irrigation, fertilizers, and pesticides.

10. Limnology

Limnology is the study of the structural and functional interrelationships of organisms of inland waters as they are affected by their dynamic physical, chemical, and biotic environments.

11. Middle /marginal feeder

Marginal feeders prefers algae and other submerged vegetation, • • And is primarily a herbivorous column feeder. • • ***Example***: • • Rohu

12. Parasitic Feeder

A parasitic feeder is an organism that feeds on another organism, known as the host, without killing it. The parasitic feeder benefits from the relationship, while the host is harmed.

Example:

- Tapeworms (parasite) feed on the nutrients absorbed by the intestines of their host (e.g., humans, animals)
- Lice (parasite) feed on the blood, skin, or feathers of their host (e.g., humans, birds)
- Leeches (parasite) feed on the blood of their host (e.g., fish, humans)

13. What are dorsal fins and their functions in locomotion

Dorsal fins are located on the back or on the top of the fish, and aid the fish in sharp turns or stops, and assist the fish in rolling. Fish may have up to three distinct dorsal fins, known as proximal, middle , and distal dorsal fins, however several fish have just two dorsal fins with the middle and distal fins fused together.

14. What is caudal peduncle?

The caudal peduncle is the base of the caudal fin. Peduncle means stem, and the caudal peduncle is where the strong swimming muscles of the tail are found. Together, the caudal fin acts like a "propeller" for the fish, and the caudal peduncle acts like a motor.

15. What do you know about fugu poisoning?/Pufferfish Poisoning

Tetrodotoxic poisoning occurs following the ingestion of pufferfish, boxfish or Porcupine fish. This is also known as puffer or fugu poisoning. The poisoning is due to toxin known as tetrodotoxin which is concentrated in the internal organs such as liver, ovary and intestines. The skin also contains some glands which secrete these toxins. Usually the body muscles are free of the toxin.

16. Examples of Herbivorous fishes

- Parrotfish
- Damselfish
- Surgeonfish

Species of these herbivorous fish include: ***parrotfish***, which eat corals that break Off the reef and excrete them as white sand; ***damselfish***, which fend off the Macro algae growth that kills coral; and ***surgeonfish***, whose most famous representative is ***Dory*** from Finding Nemo.

17. Write any three functions of ecosystem

- It regulates the essential ecological processes, supports life systems and renders stability.
- It is also responsible for the cycling of nutrients between biotic and abiotic components.
- It maintains a balance among the various trophic levels in the ecosystem.
- It cycles the minerals through the biosphere

- The abiotic components help in the synthesis of organic components that involve the exchange of energy.

18. Romer's classification (1959)

Romer's classified fishes into three classes

- ***Osteichthyes*** : includes all bony fishes:
- ***Chondrichthyes***: includes all cartilaginous fishes
- ***Placodermi***: including extinct fishes

19. Importance of taxonomy and systematics

Sound systematic data are important for conservation. Without accurate taxonomy,

- it is impossible to identify the species and evaluate their conservation status,
- it is impossible to properly manage their fisheries,
- it is impossible to evaluate the conservation value of habitats or areas,
- it is impossible to establish strategies
- it is impossible to set priorities.

20. Difference Between Bay And Estuaries

❖ **Bay:**

A bay is a body of water partially enclosed by land, connected to the open sea, with saline water. It typically has a wide mouth and is often circular or oval in shape, with water that is usually saline and similar to the open ocean. ***Examples*** of bays include the Bay of Bengal, San Francisco Bay, and Hudson Bay.

❖ **Estuary:**

An estuary is a coastal body of water where freshwater from rivers mixes with saltwater from the ocean, creating brackish water with a mixture of fresh and saltwater. Characterized by a narrow mouth and is more saline than freshwater but less saline than seawater. ***Examples*** of estuaries include the Chesapeake Bay, San Francisco Estuary, and Amazon River Estuary.

OR

Bay: A bay is a body of water partially enclosed by land, but still connected to the open sea.
Estuaries: An estuary is a coastal body of water where freshwater from rivers mixes with saltwater from the ocean, creating brackish water with a mixture of fresh and saltwater. The ***key differences*** between bays and estuaries lie in their salinity, water source, and geology, with estuaries receiving freshwater input and having a mixture of fresh and saltwater, whereas bays are primarily connected to the ocean and have more saline water.

21. Difference between lagoon and lake?

Lagoon: A lagoon is a shallow body of water separated from the ocean by a barrier, with saline or brackish water, and connected to the ocean by a narrow channel.

Lake: A lake is a body of freshwater surrounded by land, with no direct connection to the ocean.

22. Law of Tolerance with example

"The presence and success of an organism depend upon the completeness of a complex of conditions. Absence or failure of an organism can be controlled by the qualitative or quantitative. Deficiency or excess with respect to any one of several factors which may approach the limits of tolerance for that organism."

For example: The ability to exist and thrive may depend on such physical factors as dissolved oxygen, temperature, salinity, pH, etc ."Steno" refers to species with narrow tolerances (e.g.. Stenothermal= Narrow temperature tolerance) "eury" refer to those with wide tolerance (e.g. euryhaline = wide salinity tolerance)

23. Lateral line system functions

Lateral line system: Detect motion or vibrations in water

The lateral line system is a sensory system found in primarily aquatic vertebrates like lampreys, fish, and amphibians. These animals have special mechanoreceptors called lateral-line organs in their outer skin (epidermis). These organs are sensitive to small water movements, especially those created by other animals moving in the water. The lateral line system helps these aquatic vertebrates detect

changes in water pressure and movement, allowing them to sense their surroundings and potential prey or predators.

24. What is the function of bioelectric organ in fish / primary function of electric organs in electric fish?

• Bioelectric organ, system of tissues specialized for the production and use of electrical power in a living organism. • Well developed in a wide variety of fishes, both marine and freshwater, indicating an early evolutionary development, bioelectric organs probably represent a specialization of a common bioelectrical capacity of all living cells (Various other tissues and organs also possess the capacity to produce electricity—the skin of frogs and the heart, brain, and eye of higher animals including humans.) • In more than 200 fish species, the bioelectric organ is involved in self-defense or hunting. • The torpedo, or electric ray, and the electric eel have especially powerful electric organs, which they apparently use to immobilize or kill prey.

25. Skeleton of a fish

• The skeletal system forms the framework of the body and is mainly composed of bones and tissues. The main functions of the skeletal system are as follows: • Protection. • Helps in movement. • Provides shape and support. • They are the site for the storage of minerals like calcium.

• General features of fishes include fins, streamlines and scales and tails. But differences are highlighted under their skins. • Hence classification is much easier based on the skeletal system. • A variety of fishes is found in aquatic habitats Some of these fishes are cartilaginous fishes (Chondrichthyes) whereas others are bony fishes (Osteichthyes). • The skeletal system of fishes is either composed of thin and flexible cartilage or hard calcified bones or both. • They are good swimmers and their body structures are designed accordingly.

26. Name of three international shrimp / 3 commercially important species of shrimps?

Shrimps and Prawns belong to the Order Decapoda (having 10 pairs of legs: 5 pairs of walking legs and 5 pairs of swimming legs). Some of the commercially important marine shrimps are

- Pacific White Shrimp (Litopenaeus vannamei)
- Pink Shrimp (Metapenaeus dobsoni),
- Brown Shrimp (Metapenaeus monoceros), etc.

27. Ray finned fish/Why Actinoptergyii are called Ray Finned Fish?

Their fins lack muscular lobes that is why called as ray-finned fishes. • They usually possess swim bladders, gas filled sacs along dorsal wall of the cavity that regulate buoyancy. • Paired fins supported by dermal rays • Base of fins is not muscular • Tail fins almost equal to upper and lower lobes • Blind olfactory

Ray-finned fishes include herbivores, carnivores, omnivores, zooplankton (for example, small crustaceans such as copepods and fish larvae) consumers, and detritus feeders, or those that feed on dead organic matter.

28. Common Carp

• Common carp ***Cyprinus carpio*** was introduced from Thailand to Pakistan in 1964 for the purpose of aquaculture. • Due to its high tolerance to temperature and turbidity, and prolific pond breeding habit, it was established promptly in most of natural inland waters, including rivers, lakes, streams, canals, wetlands and even village ponds of the country. • Although common carp became one of the most abundant cyprinid species in inland waters and important food fish in Pakistan, its impact is not well documented.

29. Feeding Habit of carps

Carps are omnivorous fish, meaning they eat both plant and animal matter. Their diet typically includes algae, aquatic plants, insects, crustaceans, and small fish. Carps are bottom-feeders, so they often scavenge for food at the bottom of ponds, lakes, or rivers. They have specialized pharyngeal teeth that help them crush and grind their food. Carps are known to be opportunistic feeders, consuming a wide variety of food sources depending on availability.

30. Write characteristics of carp

• They have a greater tolerance of low oxygen levels, pollutants and turbidity than most native fish, and are often associated with degraded habitats, including stagnant waters.

• Changes to water flows, declining water quality and other changes to river habitats over the past few decades have negatively affected many native fish while favoring carp.

• Body bilaterally symmetrical, moderately elongate, its dorsal profile more arched than the ventral profile.

• Head without scale, body with cycloid scale.

31. Feeding of pf herring like fish?

Herring-like fish, such as herring, sardines, and anchovies, are typically planktivores or filter feeders, feeding on:

- Phytoplankton (microscopic plants)
- Zooplankton (microscopic animals)
- Small crustaceans (such as krill or copepods)
- Small fish larvae

They use their small teeth and gill rakers to filter the water and trap their food. Some species may also feed on detritus or algae.

32. Feeding behavior of pf Herring fish

Filter feeding: using their gill rakers to strain the water and capture small particles

Surface feeding: feeding on the surface of the water, often in large schools

Mid-water feeding: feeding in the middle of the water column, often on small fish or crustaceans.

33. Marine Fisheries, with example?

Marine fisheries are conducted in all the oceans and seas of the world, including bays and estuaries . The capture and culture of aquatic organisms in salt water accounts for the bulk of the fishery products that reach world markets. Marine fishery deals with the fishery aspects of the sea water or ocean. ***Example:*** Hilsa migrates from the sea to the river for breeding. Dr. Hora studied the migration of Hilsa.

34. How freshwater fishes regulate their internal salt concentration?

Freshwater fishes live in water with a much lower concentration of salts than they require inside their bodies. As a result, osmosis tends to promote the absorption of water into their bodies. To counteract this, their kidneys efficiently excrete water, and their gills reabsorb salt from the water. This helps maintain a stable internal environment and prevents excessive water uptake.

35. Brackish Water Aquaculture

Brackish water areas are rich in oxygen and plankton. Brackish water organisms such as milkfish (Chanos chanos), Lates calcarifer, Mugil sp. prawns, mussels, etc. can be cultured in various ways such as in tidal ponds, pens, cages or by rack, raft or rope culture (mainly in case of oysters). Mangrove crab, Scylla serrata, are also cultured in the shallow brackish water regions by constructing bamboo cages.

LONG QUESTIONS(5)

1. J.Muller's classification (1844)

• Earliest classification was proposed by J. Muller (1844) who proposed six subclasses in class Pisces.
• ***Class PISCES Subclasses:***

1) DIPNOI (Lung fishes);
2) TELEOSTEI (Bony fishes);
3) GANOIDEI (Polypterus, Amia, Lepidosteus);
4) ELASMOBRANCHI (Cartilagenous fishes);
5) MARSIPOBRANCHI (Cyclostomes);
6) LEPTOCARDII (Amphioxus).

2. Adaptation of gills for efficient gas exchange

1. Thin (Flattened cell) • Short diffusion distance

2. Permeable membranes • Allow gases to diffuse

3. Good vascular supply • Maintain concentration gradient

4. Large surface area • Allow more diffusion of gases

3. Types Of Caudal Fin

The caudal fin is known commonly as the tail fin. It is the primary appendage used for locomotion in many fishes. The main types of Caudal fins found in bony fish are:

- **Homocercal shape tail:** it is the most common caudal fin shape, shared by most fishes. it includes square, truncate, slightly forked and deeply forked types.
- **The heterocercal tail:** Possessed by only a few primitive fishes such as stugeron,
- paddlefish and sharks. The caudal peduncle is the base of the caudal fin. Peduncle means stem, and the caudal peduncle is where the strong swimming muscles of the tail are found. Together, the caudal fin acts like a "propeller" for the fish, and the caudal peduncle acts like a motor.
- **Rounded tail:** Slow swimming, accelerating, and maneuvering
- **Truncated (triangular) tail:** Turning quickly
- **Lunate (moon shaped) tail:** Continuous long distance swimming
- **Forked tail:** Rapid swimming, somewhat sustained with bursts of speed

4. Difference between Actinopterygii and Sarcopterygii.

❖ **Subclass Actinopterygii**

• (Ray-finned fish) • Their fins lack muscular lobes that is why called as ray-finned fishes. • They usually possess swim bladders, gas filled sacs along dorsal wall of the cavity that regulate buoyancy. • Paired fins supported by dermal rays • Base of fins is not muscular • Tail fins almost equal to upper and lower lobes • Blind olfactory • ***Examples*** • Ray fined fishes

❖ **Subclass Sarcoptergyii**

• (Lobe-finned fish) • Have muscular lobes associated with their fins and usually use lungs for gas exchange. • Pneumatic sacs function as lungs or swim bladder • Paired fins with muscular lobes • Some bony fishes are hermaphrodites and a number of species exhibit parthenogenesis • Fertilization is usually external • Development is usually oviparous (egg laying) but can be ovoviviparous or viviparous • ***Examples*** • Lungfishes, coelacanths

5. Biological Rhythm

We are strongly controlled by internal (endogenous) circadian rhythms affecting temperature, physiology and behavior etc. • Physiology and behavior are strongly controlled by such rhythms, with periods of a year, a lunar month, a day or a tidal cycle. The reproductive biology and feeding patterns are intimately linked to such rhythms.

• ***Example*** include vertical feeding migrations from deep water towards the surface to feed on plankton, annual migrations from feeding to breeding grounds, synchrony of breeding on spring high tides and movement from the bottom to the middle of the water column on high and low tides.

6. What is Ciguatera poisoning?

• Ciguatera poisoning is due to the consumption of fishes contaminated with ciguatoxin, a toxin which acts as a sodium channel agonist. • This toxin is produced by a microscopic marine dinoflagellate Gambierdiscus toxicus and through the food chain it reaches the fish, mainly the reef fishes. • The toxin accumulates mainly in the liver and testis of the fish as well as other viscera, while the flesh (muscle) of the fish is much less affected. • Cooking or storage of the fish has no effect on the toxicity. • Some of the fishes which have been implicated with ciguatera poisoning are groupers, barracuda, amberjack, sturgeon fish.

7. What are Lobsters?/Lobster limnology and its scope in study

Lobsters are one of the most valuable and highly priced crustaceans as well as an important export commodity. • The massive head bears five pairs of walking legs; the abdomen is well developed, muscular and bears five pairs of swimmerets. Lobsters are solitary animals that crawl on the sea floor and hide under rocks. • Among the 12 species recorded from Indian coastal waters, only ***four species of Spiny Lobsters*** (Panulirus homarus, P. ornatus, P. polyphagus, P. versicolor) and ***one species of Slipper Lobster*** (Thenus orinetalis) are commercially important.• Growth is slow, most lobsters take more than 3 years to attain maturation and they have a prolonged life-cycle. Grow-out and fattening of lobsters is commonly done in indoor tanks and cages to fetch better price in market.

8. Composite Fish Farming

Composite Fish Farming: • It is found that if few selected species of fish are stocked together in proper proportion in a pond, total production of fish is increased many times. • This mixed farming is called composite farming. • It has some advantage-compatible species do not harm each other, all available areas are fully utilised, no competition among different species is found and fish may have beneficial effect on each other. • Catla catla, Labeo-rohita and Cirrhina mrigala are surface feeder, column feeder and bottom feeder respectively and are used for composite farming.

9. Pyloric caeca and stomach function in digestive system of fish

• The stomach varies greatly in fishes, depending upon the diet. • In most predacious fishes it is a simple straight or curved tube or pouch with a muscular wall and a glandular lining. • Food is largely digested there and leaves the stomach in liquid form.

The junction between the stomach and the intestine is marked by a muscular valve. • Pyloric ceca (blind sacs) occur in some fishes at this junction and have a digestive or absorptive function or both. • The intestine itself is quite variable in length, depending upon the fish's diet. • It is short in predacious forms, sometimes no longer than the body cavity, but long in herbivorous forms, being coiled and several times longer than the entire length of the fish in some species of South American catfishes.

10. Difference between lotic and lentic freshwater ecosystem.

❖ **Lotic Freshwater Ecosystem:**

• In simple terms, the water bodies moving in one direction is known as a lotic freshwater ecosystem. Rivers and streams are common examples of lotic ecosystems. • There are many rivers and stream that flows from their origin and ultimately meets with other water channels or oceans at its mouth. • Lotic freshwater travel through different locations from its source to mouth. •

❖ **Lentic Freshwater Ecosystem:**

• An aquatic ecosystem within stagnant or still water like ponds and lakes is known as Lentic Freshwater Ecosystem. • Lentic ecosystem found in various sizes ranging from a few square meters to thousands of square km. • Some ponds last just for a few months as these are seasonal like sessile pools. • On the other hand, lakes may exist for many years. Lentic ecosystem, i.e. Ponds and lakes, support a very limited number of species.

11. Fresh Water & Marine Ecosystem

❖ **Freshwater Ecosystem:**

The freshwater ecosystem is an aquatic ecosystem that includes lakes, ponds, rivers, streams and wetlands. These have no salt content in contrast with the marine ecosystem. The freshwater ecosystem is mainly divided into three types based on its region : Lotic, lentic, and wetland freshwater ecosystem.

❖ **Marine ecosystem:**

The marine ecosystem includes seas and oceans. These have a more substantial salt content and greater biodiversity in comparison to the freshwater ecosystem.

A marine ecosystem is a dynamic and interconnected community of living organisms, including tiny plankton, massive blue whales, and everything in between. These ecosystems cover over 70% of the Earth's surface and encompass a vast range of habitats, from shallow tide pools to the deepest ocean trenches. Marine ecosystems provide essential services, including primary production, food webs, nutrient cycling, coastal protection, and climate regulation, supporting an incredible array of biodiversity. They are crucial for human well-being, providing half of the oxygen we breathe, a significant source of food, and livelihoods for millions of people.

12. Name pathogenic and parasite that causes diseases in fishes?

Fish Diseases Caused by Parasites and Pathogens:

1. Bacterial Diseases: Two bacterial diseases are very important.

- Abdominal dropsy of Carps is caused by Aeromonas punctuata.
- Furunculoris of Salmons and trout's is caused by Aeromonas salminicida

2. Viral Diseases: Economically most important is the viral haemorrhage septicaemia (VHS) of rainbow trouts

3. Protozoan Diseases: Main protozoan diseases are caused by Costia, Myxobolus and Trypanosoma

4. Fungal Diseases: The gill rot (branchyomyces) of carps involves the attack of Saprolegnia on the gills of carps.

5. Worm Diseases: Worms of four groups are parasites on fish. The flatworms (trematodes), tapeworms (cestodes), round worms (nematodes) and thorny-headed worms (acanthocephalans).

6. Common Ectoparasites: Two ectoparasites of fish are most important, the fish lice (Argulus, Lernaea and Ergasilus) and the fish leech (Piscicold). Both parasites weaken fish by feeding on its blood

13. Lamprey And Its Habitat

• Lampreys generally inhabit shallow coastal regions and freshwater environments. • They feed on small invertebrates or larger fish species by attaching to the body wall and sucking their blood. • There are both freshwater and marine species of lamprey. • However, marine species must migrate to freshwater in order to spawn and for their young larvae to develop. • Although there are native freshwater lamprey species found in North America, sea lampreys have been invading the Great Lakes over the past 100 years. This invasion was the result of man-made canals, built in the early 1900s, through which the sea lampreys were able to reach the lakes. • They have caused a great deal of destruction to the fish industry by preying on economically important populations of trout and other fishes. • Since they do not have natural predators in the Great Lakes, it has been a challenge for the U.S. and Canadian governments to control their populations. • Methods such as chemically sterilizing males or using pheromones to disrupt their life cycles are currently being investigated. • Although they are a nuisance in the Great Lakes of North America, there are many countries that value lampreys as a delicacy. Countries like Portugal, Spain, France, Sweden, Finland, South Korea, and the Baltic nations consume considerable amounts of lamprey, and populations have actually decreased in some of these regions due to overfishing

14. Classification of fishes on the basis of feeding habitat

Classification of fish on the basis of feeding habitats . Basic eating groups among fish are:

1) ***Carnivores*** : Fish exclusively consume animal matters as food (e.g., sharks, barracudas).
2) ***Herbivores***: Fish exclusively consume plant matters as food (e.g., parrotfish, surgeonfish).
3) ***Omnivores***: Fish consume both plant and animal matters as food (e.g., tilapia, catfish).
4) ***Larvivores***: Fish exclusively consume insect larvae as food (e.g., some species of trout).
5) ***Planktivorous***: Fish exclusively consume plankton as food. (e.g., sardines, anchovies).
6) ***Voracious***: Fish consuming or craving large quantities of food (e.g., piranhas, groupers).
7) ***Detritovores***: Fish consume decaying plants and animals.(e.g., some species of catfish).
8) ***Cannibalistic***: Cannibalism is the act of consuming another individual of the same species as food. (e.g., some species of cichlids).
9) ***Piscivores*** : A piscivore is a carnivorous animal that eats primarily fish (e.g., pike, bass).
10) ***Carcinophagus:*** Fish that primarily consume crustaceans, such as crabs and shrimps, as food (e.g., some species of wrasses, eels).

15. Characteristics of Fish

General characteristics:

➢ Fish are aquatic, gill-bearing animals

➢ that lack limbs with digits.

➢ Most fish are cold-blooded,

➢ allowing their body temperatures to vary as ambient temperatures change.

➢Fish share with vertebrate a basic body plan of bilateral symmetry with a dorsal supporting structure the notochord as a precursor the series of harder vertebrae,

➢Their body is divisible into head, trunk and tail. Neck is absent.
It has a spindle shaped body. It is helpful in swimming.

➢Respiration is by gills. Gills are the extensions of the pharynx. In the elasmobranches fishes, the gills will open separate. In bony fishes the gill slits are covered by operculum.

➢On the head a pair of nostrils are present internal nostril are absent. In Dipnoi Internal nostrils are present

16. Fish mouth morphology

❖ **Mouth Type of fish**

(i) ***Long and skinny bill*** •Tweezer like • Poking into crevices
(ii) ***Downward orientation*** • Oriented mouth • Useful to suck food from bottom • E.g., Janitor fish
(iii) ***Large mouth*** • Tearing large prey
(iv) ***Beaklike mouth*** • -used to graze on small algae growing on hard surfaces • E.g Parrot fish

(v) ***Upward orientation*** • Used to capture prey on water surface

❖ **Mouth Position of fish**

Most fish mouths fall into one of three general types:

(i) ***Superior***, or sometimes called ***supra-terminal, mouths*** are upturned. • The superior mouth is oriented upwards, and the lower jaw is longer than the upper jaw. • Archers, half-beaks, and hatchet fish are all examples of species of aquarium fish that have a superior mouth.

(ii) ***Terminal mouths*** point straight forward and are the most common mouth type. Terminal mouths are located in the middle of the head and point forward. Both jaws are the same length. More fish have this mouth type than any other.• Most barbs, cichlids, gouramis, and tetras have terminal mouths.

(iii) • ***Inferior***, ***or sub-terminal, mouths*** are turned downward. • Also called a sub-terminal or ventral mouth, the inferior mouth is turned downward. The lower jaw is shorter than the upper jaw, and the jaw will often be protrusible. • Most members of the catfish family have inferior jaws, and many of them also have a sucker mouth as well.

(iv) ***Protrusible mouth*** • A protrusible mouth allows a fish to extend its reach when attempting to snatch prey or food particles. • For example, kissing gourami uses its protrusible mouth to defend territory against others of the same specie.

17. Note on Class MYXINI

• Considered as most primitive fishes

• Known for its slimy secretions

• Cartilaginous skeleton

• Hagfish have elongated,

• eel-like bodies, and paddle-like tails.

• Sucker like mouth

• Mouth with four pairs of tentacles

• Five to fifteen pairs of pharyngeal slits

• Olfactory sacs apart to mouth

• Live in mud and sand of marine water.

• Hagfish are typically about 50 cm (19.7 in) in length.

• They are scavengers

• (enter in the large fish through mouth, eat internal parts leaving on skin and bones)

• Reproduction: rarely known

Example • Hag fishes

18. Characteristics of Osteichthyes

• Includes all bony fishes ***Example***: Bony fishes

• With more than 29,000 species of bony fishes, it is the largest class of vertebrates.

• It includes both marine and freshwater fishes, most of the commercially used fishes are included in this class.

• The size ranges from less than 8 mm in Paedocypris progenetica, which is also the smallest known vertebrate to 4 m and weigh about 1500 kg in the ocean sunfish (Mola mola).

• Their endoskeleton is made up of bones.

• Bony skeleton

• Operculum cover gills opening

• 20,000 species

• Present in Freshwater and saltwater

• Pneumatic sacs functions as lungs or swim bladder

• Bone in their skeleton, bony operculum covering the gills openings, lungs or swim bladder.

• Lateral line system

Fins: Dorsal, pectoral, caudal, pelvic, anal.

• Paired and median fins are present, which are supported by long rays of cartilage or bone. Fleshy lobed fins are present in sarcopterygians.

• These types of fins are supported by bones having joints.

• These are adapted to live at the bottom of the sea.

• First fossils of bony fishes are from late Silurian deposits.

19. Characteristics of chimaeras

Chimaera, (subclass Holocephali), any of numerous cartilaginous fishes related to sharks and rays in the class Chondrichthyes but separated from them as the subclass (or sometimes class) Holocephali. • Like sharks and rays, chimaeras have cartilaginous skeletons, and the males possess external reproductive organs (claspers) derived from the pelvic fins and used to introduce sperm into the body of the female. • Unlike sharks and rays, chimaeras have a single external gill opening, covered by a flap as in the bony fishes, on each side of the body. • Male chimaeras, unique among fishes, also possess a supplemental clasping organ, the tentaculum, on the forehead and in front of each pelvic fin.

20. Methods to reduce Turbidity of water

Here are the methods to reduce turbidity in water:

- **Sedimentation**: Allow particles to settle to the bottom.
- **Coagulation**: Add chemicals to remove dirt and impurities.
- **Filtration**: Pass water through filters, such as sand or membranes.
- **Centrifugation**: Use centrifugal force to separate particles.
- **Flocculation**: Add chemicals to form larger particles for easier removal.
- **Ultraviolet (UV) light:** Disinfect and clarify water.
- **Nanofiltration/Ultrafiltration:** Use membranes with small pores to remove particles.
- **Activated Carbon Filtration:** Remove impurities and improve water clarity.

21. What is the purpose of Aquaculture & Characteristics of Aquaculture

• The word aquaculture combines the Latin aqua-, “water,” with culture, also from a Latin root, meaning “agriculture” or “a cultivating.” • Originally, this term referred only to fish farming. • Aquaculture is breeding, raising, and harvesting fish, shellfish, and aquatic plants. • Basically, it's farming in water.

Aquaculture may be defined as the “***farming and husbandary of economically important aquatic animals and plants under controlled conditions.”***

PURPOSE:

- To increase fish production and reduce dependence on capture fisheries
- To provide a viable alternative to traditional fishing methods
- To produce food and products in a controlled and sustainable manner
- To help rebuild stocks of threatened or endangered species
- To create healthier habitats and promote environmental stewardship
- To produce industrial products such as seaweed and algae for human consumption
- To raise shellfish such as oysters and scallops for human consumption
- To farm aquatic plants and animals for research and educational purposes.

22. Characters of Aquaculture /Benefits of aquaculture?(Any Five)

High Productivity: In comparison to agriculture or veterinary practices, aquaculture has been found to be more productive.

Integrated Farming: Aquaculture with agriculture, horticulture or animal husbandary is found to be more profitable,

Recycling: Aquaculture gives efficient means of recycling domestic and/or agriculture wastes and thus, helps in protecting our environment

Rural Development: Aquaculture helps to integrate rural development by generating employment opportunities and would thus help to arrest the exodus of population from rural to urban areas.

Intensive Fish Culture: In aquaculture practice, owing to the fishes three-dimensional utilization of the water body can be crowded more closely (200/ m3) and grown through water recirculation system. This gives high yield of about 25 t/ha/year

Earning Foreign Exchange: • Commercially important items, such as ornamental fishes, Artemia cysts, prawns, lobsters, crabs, frog legs, etc., produced through aquaculture are highly valued and can earn good foreign exchange.

Ranching: • Fish seeds and larvae of economically important fishes of capture fisheries are artificially recruited in fish hatcheries through aquaculture, called ranching or aquarange farming. It is commonly known as “culture based capture fishery” and involves restocking of the wild stock for harvesting. This process has replenished the dwindling stock of rivers and seas.

Replenishment of Coral Reefs: • Coral reefs that have been destroyed naturally or anthropogenically are being replenished through construction of artificial coral reef. •

Creating Leasure-Time Activities: • Leasure-time activities can be maintained through sport fishing and creating home and public aquaria.

Mariculture and Fisheries Enhancement: • In many reef areas the use of mariculture is fast growing. It provides an alternative source of income and employment in coastal areas. Replenishment schemes of species that has been greatly diminished has been undertaken in certain areas. This has been useful for returning species such as giant clams and trochus to reefs where they have been largely or completely exterminated by overfishing.

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